

Overview

This page reviews research and practical lessons on **public-transport kiosks and stop-level information displays** and translates them into a buildable TransitSense kiosk design.

Across the literature, one theme repeats:

1. Passengers value **certainty and clarity** more than feature richness.
2. Real-time displays reduce anxiety and perceived waiting time.
3. Trust collapses when predictions look precise but are stale or wrong.
4. Interactive kiosks are useful, but only if placement and simplicity are right.
5. Physical wayfinding and digital information must support each other.

Core sources reviewed

1. Real-time information systems for public transport: effects on customers
2. Real-time bus arrival information system: an empirical evaluation
3. Interactive Displays in Public Transport – Challenges and Expectations
4. Implementation of Interactive Transit Information Kiosks at New York City Transit Facilities
5. Mind the Gap: Navigating the Space Between Digital and Physical Wayfinding in Public Transit

Note: Some sources were available mainly through abstracts or secondary summaries rather than full open-text technical details. The conclusions below stay cautious and focus on consistent patterns across sources.

What the literature consistently says

1. The main job of a kiosk is to reduce uncertainty

Research on real-time passenger information consistently shows that passengers benefit when the system tells them:

- whether the bus is actually coming,
- how long the wait is,
- whether service is delayed,
- and whether they should change behavior.

A kiosk is valuable not because it is flashy, but because it reduces the psychological cost of waiting.

2. Accuracy without freshness is not enough

The empirical evaluation literature shows that real-time systems can systematically under- or over-estimate waiting times. Once users notice this, trust drops quickly.

That means a good kiosk should never show ETA alone. It should also show:

- **last updated** time,

- stale/live status,
- and graceful fallback wording when predictions are uncertain.

3. Simpler displays usually win at stops

Interactive kiosk studies and public-display design work suggest that many passengers do not want to navigate deep menus while waiting. They usually want a small number of answers immediately visible.

For TransitSense, the stop kiosk should therefore prioritize:

1. next arriving buses,
2. ETA,
3. crowd level,
4. route direction / destination,
5. service alert / unavailable-live-data message.

Interactive features can exist, but they should not be the core dependency.

4. Placement and visibility matter as much as UI

A well-designed interface still fails if passengers cannot spot or read it.

Common practical lessons:

- keep the screen visible from typical waiting positions,
- use large text readable in daylight,
- avoid overloading the display,
- separate critical live information from secondary guidance,
- and make the stop identity unambiguous.

5. Digital and physical wayfinding must reinforce each other

The wayfinding literature emphasizes that digital guidance does not replace the physical environment.

A kiosk works best when it clearly matches:

- the stop name,
- route numbers,
- destination names,
- direction of travel,
- and nearby landmark cues.

This is especially important for first-time users.

Method / design families

A. Passive real-time arrival board

Problem

Provide immediate next-bus information to all waiting passengers.

Data

- ETA feed
- route / stop metadata
- optional crowd status
- service alerts

Method

A simple digital display continuously cycles or refreshes live arrival rows.

Results in practice

This is the most robust and understandable type of kiosk. It supports glanceable use and does not require user interaction.

Constraints

- Limited interaction depth.
- Must keep screen uncluttered.
- If ETAs are unreliable, the display can damage trust.

Privacy stance

Very strong. No personal data required.

Hardware cost level

Low to medium depending on display type.

Connectivity assumptions

Can tolerate weak networks if the display caches last-known data and clearly marks staleness.

Fit for TransitSense

Best MVP choice.

B. Interactive trip-planning kiosk

Problem

Help users explore routes, destinations, and alternatives.

Data

- route map
- stop network
- arrival feed
- service alerts
- nearby places / landmarks

Method

Touchscreen kiosk with menu flows, map interaction, and search.

Results in practice

Useful in large interchanges or unfamiliar systems, but adoption depends heavily on placement, signage, and user awareness.

Constraints

- More software complexity.
- More screen-state design required.
- Slower for users who only need a quick answer.

Privacy stance

Strong if it does not store personal queries.

Hardware cost level

Medium.

Connectivity assumptions

Needs local fallback pages to remain useful offline.

Fit for TransitSense

Good future extension, weaker first MVP fit.

C. Hybrid kiosk: glanceable board + simple interaction

Problem

Balance fast live information with limited wayfinding help.

Data

- ETA feed
- occupancy feed
- route / direction data
- static route map
- alerts

Method

Default screen behaves like a passive arrival board. A simple tap or button can switch to:

- route map,
- stop sequence,
- nearby interchange guidance,
- or service-status details.

Results in practice

This is often the most balanced design because it preserves quick readability while still supporting deeper guidance.

Constraints

- Needs disciplined screen hierarchy.
- Interaction must be optional, not required.

Privacy stance

Strong.

Hardware cost level

Low to medium.

Connectivity assumptions

Good if static assets are bundled locally and only live values depend on the network.

Fit for TransitSense

Best overall design direction.

Comparison table

Design principles for TransitSense

Recommended default screen

The kiosk home screen should answer, at a glance:

1. Which buses are coming next?
2. How long until they arrive?
3. How crowded are they?
4. Is the live data fresh?

A strong default row could be:

- 12A → Central Station — 6 min — Moderate crowd
- 21C → Market Road — 11 min — Standing room
- 5B → Live update delayed — Last updated 2 min ago

Recommended secondary information

Allow optional access to:

- route map,
- stop sequence,
- nearby interchange / transfer note,
- simple landmark-based guidance,
- service alerts.

Accessibility and readability rules

- Large, high-contrast text.
- Avoid small dense maps on the home screen.
- Use consistent route colors or symbols.
- Keep route number and destination visually dominant.
- Provide a clear stale-data state.

Constraints and risks to make explicit

1. The kiosk is only as trustworthy as the ETA feed.
2. Too much interactivity hurts glanceability.
3. Weak network must not produce blank screens.
4. Physical placement affects usability heavily.
5. Stop identity and route direction must never be ambiguous.

TransitSense recommendation

Default choice

Build a **hybrid kiosk** whose home screen behaves like a passive arrival board and whose deeper pages provide only simple, optional wayfinding.

What to emphasize in the SIH pitch

- lower passenger uncertainty,
- crowd-aware waiting decisions,
- offline-first display behavior,
- accessibility and first-time-user friendliness.

Avoid for first MVP

Do not make the kiosk dependent on:

- complex trip planning,
- heavy map interaction,
- or always-online behavior.

These can be shown as extensions, but the MVP should stay focused and reliable.